

4. (a) Welsh Water is responsible for supplying and maintaining safe water supplies in Wales. It monitors the composition of our drinking water.

The table below shows the composition of stream water supplying a reservoir in Wales. The samples were taken from three different streams.

Component	Concentration (units)		
	stream 1	stream 2	stream 3
sodium ions (Na^+)	5.0	5.0	5.0
potassium ions (K^+)	6.3	6.2	6.2
magnesium ions (Mg^{2+})	15.3	15.4	15.5
calcium ions (Ca^{2+})	50.4	50.1	50.1
chloride ions (Cl^-)	30.1	30.0	30.2
dissolved sulfur dioxide	6.0	8.2	10.0
dissolved carbon dioxide	0.6	0.9	1.6
pH	5.7	5.5	5.4

Use the information in the table to answer the following questions.

- (i) State the name of **one** ion whose concentration is the same in each stream. [1]

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- (ii) There is more dissolved sulfur dioxide in stream 3 than in stream 1.

I. Use the equation:

$$\% \text{ difference} = \frac{\text{concentration in stream 3} - \text{concentration in stream 1}}{\text{concentration in stream 3}} \times 100$$

to calculate the percentage difference between the dissolved sulfur dioxide in stream 1 and stream 3. [2]

percentage difference =



- II. Complete the following sentence by underlining the correct term in the bracket.

An increase in dissolved sulfur dioxide makes the rainwater (**more acidic / become neutral / less acidic**). [1]

- III. State how an increase in dissolved sulfur dioxide affects the pH of water. [1]

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- (iii) A technician believes that streams 1, 2 and 3 have water of the same hardness. Use the data in the table to explain whether you agree with the technician. [3]

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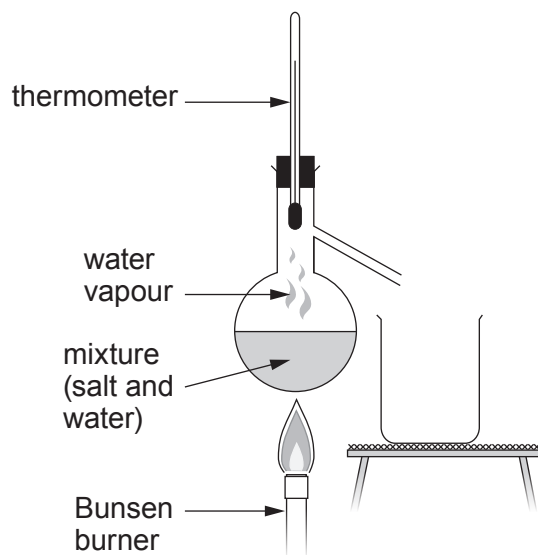
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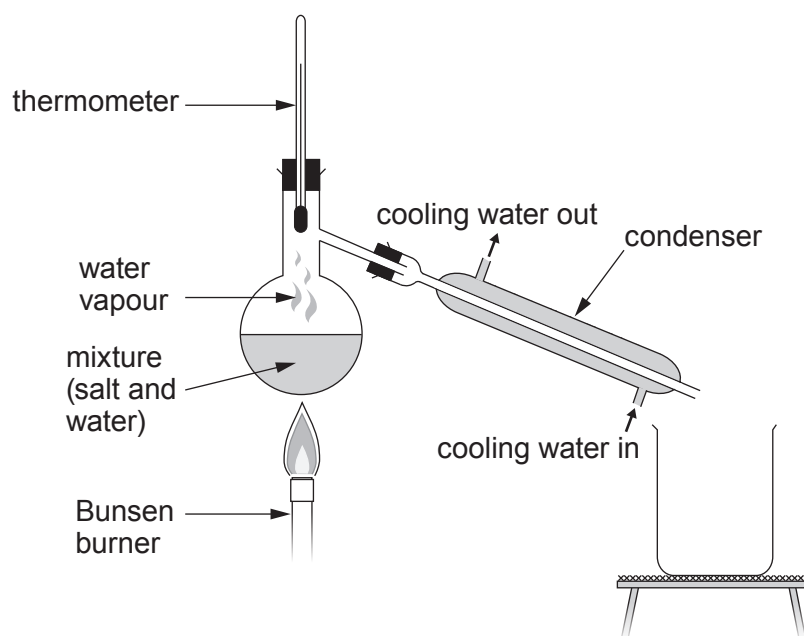
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- (b) When water is in short supply, seawater can be desalinated. In a school laboratory, distillation apparatus can be used to desalinate seawater. A student was given the choice of apparatus A or B.



Apparatus A



Apparatus B



- (i) State the temperature that the thermometer will record during the desalination of water. [1]

..... °C

- (ii) A student believes that she will collect more water using apparatus A than apparatus B.

Explain whether you agree. [2]

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- (iii) South East England needs an alternative source of water to supply drinking water. One of the options is to build a desalination plant.

State **one** disadvantage of desalination of sea water. [1]

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